Chapter 4: Loops and Iteration

Python Programming Study Guide Series

ITERATIVE STATEMENTS

Loops execute a block of code multiple times until a specified termination condition is satisfied.

For Loop

Iterates over items of sequence or range.

```
for x in range(1, 4): print(x)
```

While Loop

Repeats as long as condition evaluates to True.

```
count = 0 while count < 3: count += 1
```

LOOP CONTROL DIRECTIVES

- `break`: Instantly exits the innermost active loop.
- `continue`: Skips remaining commands in current iteration and re-evaluates the loop condition.
- `pass`: Syntactic placeholder used when no code action is required.

```
# Example: Loop Controls numbers = [10, 15, 20, 25, 30] for n in numbers: if n == 15: continue # Skip 15 if n == 25: break # Stop at 25 print(n) # Output: 10, 20
```